DEIS Lab 2: Basic Robotics 2
DT8007 VT26, Halmstad University

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1 October 2026

Introduction

In this lab we programmed an Arduino UNO R4 WiFi, did some basic hardware prototyping, flew a Ryze Tello drone (manually and from Python), and got a SparkFun RedBot ground robot to move.

1 Arduino

1.1 Getting started

We installed the UNO R4 board package in the Arduino IDE, selected the board and port, and uploaded the Blink example. We ran it on a classic Arduino Uno as well as on the UNO R4 WiFi.

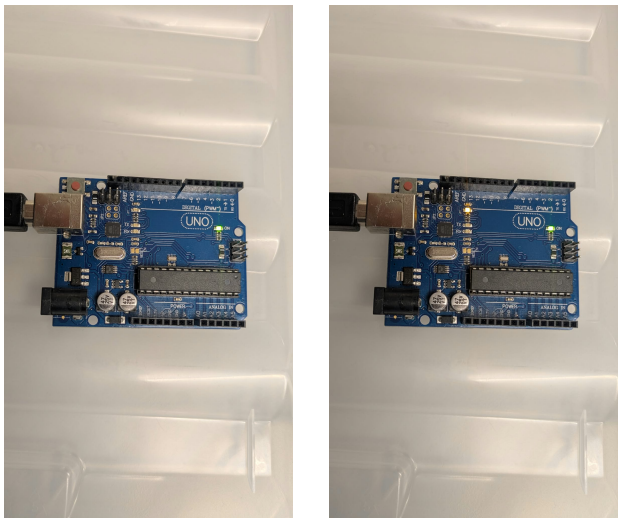


Figure 1: Blink example on an Arduino Uno: LED L off (left) and on (right).

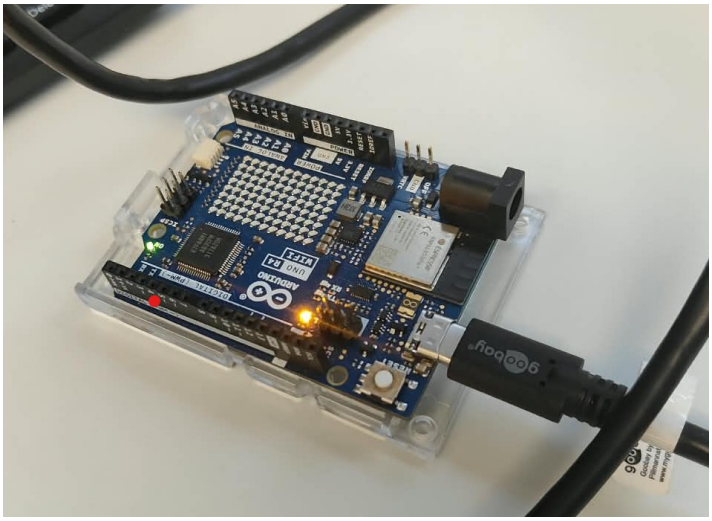


Figure 2: Sketch running on the Arduino UNO R4 WiFi (LED L on).

1.2 Connecting to WiFi

We uploaded the `SimpleWebServerWiFi` example with the course router's SSID and password, opened the IP address from the Serial Monitor in a browser, and turned the LED on and off (`/H` and `/L`).

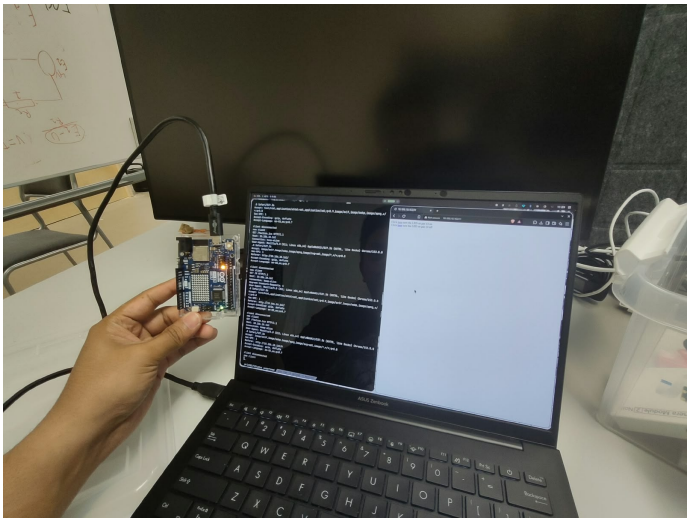


Figure 3: LED on pin 13 turned on from the browser over WiFi. The terminal on the left shows the HTTP requests received by the board.

2 Prototyping

Together with the supervisor we did simple tasks with a multimeter, soldering iron and drill. We used the multimeter to check batteries, and soldered LEDs and wires onto a perfboard held in a helping-hands stand.



Figure 4: Multimeter set up for checking batteries.

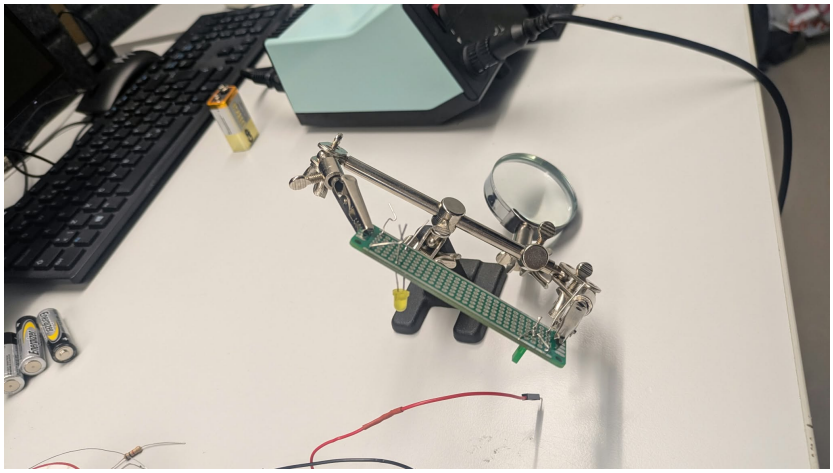


Figure 5: Perfboard with an LED in the helping hands, next to the soldering station.

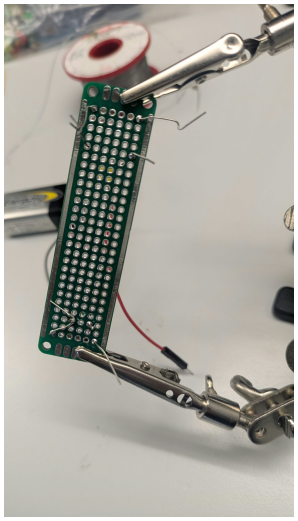


Figure 6: Solder joints on the perfboard.

3 Drone

3.1 Manual flying

Using the Ryze Tello app we took off, rotated left and right, flew up and down, forward and backward, strafed left and right, and landed.

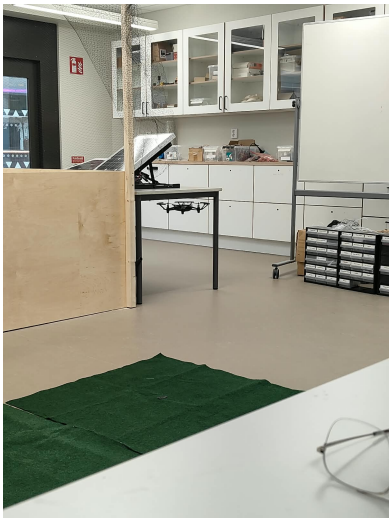


Figure 7: Tello hovering in the lab during manual flight.

3.2 Control via Python

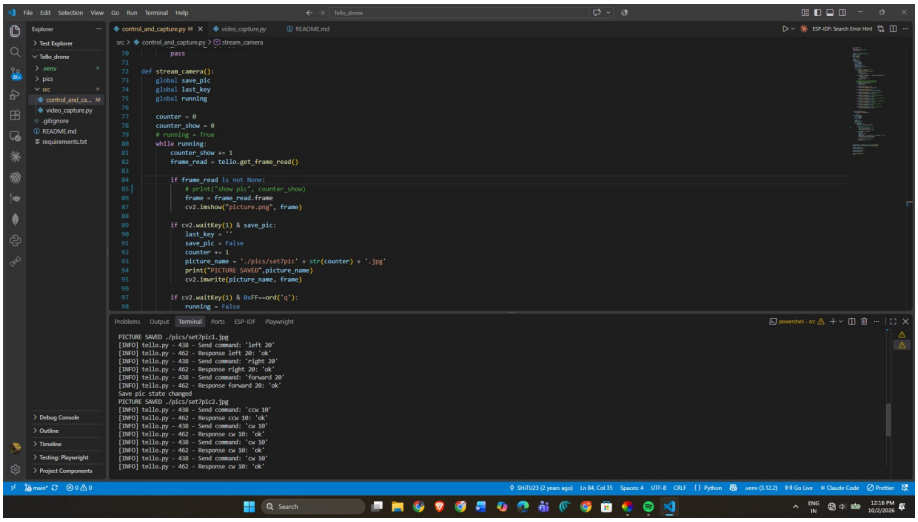


Figure 8: Controlling the drone from Python. The terminal shows the commands sent, the responses, and the saved camera frames.

We then added a simple autonomous sequence that only rotates: take off, wait 1 s, rotate 90° clockwise, wait 1 s, rotate 90° counter-clockwise, wait 1 s, and land.

```
========== Initiating Autonomous Action 1 (T0=W=R90=W=R(-90)=L) ==========
1. Initiating Take off.......
[INFO] tello.py - 438 - Send command: 'takeoff'
[INFO] tello.py - 462 - Response takeoff: 'ok'
====== Take off done =====
2. Wait 1 second........
3. Rotate 90 degree clckwise........
[INFO] tello.py - 438 - Send command: 'cw 90'
[INFO] tello.py - 462 - Response cw 90: 'ok'
====== Rotation complete ======
4. Wait 1 second.......
3. Rotate 90 degree Anti-clckwise........
[INFO] tello.py - 438 - Send command: 'ccw 90'
[INFO] tello.py - 462 - Response ccw 90: 'ok'
====== Rotation complete =====
5. Wait 1 second........
6. Initiating Landing.........
[INFO] tello.py - 438 - Send command: 'land'
[INFO] tello.py - 462 - Response land: 'ok'
===== Landing done =====
break the loop
```

Figure 9: Output of the autonomous sequence (take off, rotate, land).

4 Ground robot

We assembled the RedBot (main board, IR line sensors, wheel encoders, alkaline battery pack) and worked through the SparkFun tutorials 1–3 (basic motions), 6 (line following) and 7 (encoders)[1].

Comment. We did not have black tape to make a line-following track, so we connected the RedBot to the Arduino UNO R4 WiFi over serial and drove it from a browser instead. Commands go from WiFi to the UNO and from the UNO to the RedBot:

$$\text{Browser} \xrightarrow{\text{WiFi}} \text{UNO R4 WiFi} \xrightarrow{\text{serial RX/TX}} \text{RedBot}$$

The UNO TX (pin 1) is wired to the RedBot RX (A1), the UNO RX (pin 0) to the RedBot TX (A0), and the grounds are connected; the link runs at 9600 baud. The UNO creates its own WiFi network

[1] https://learn.sparkfun.com/tutorials/experiment-guide-for-redbot-with-shadow-chassis

and serves a page with W/A/S/D controls. Each key press reaches the UNO as an HTTP request, and the UNO passes the letter on to the RedBot over serial. The RedBot drives the motors and replies with its IR line-sensor and encoder readings, which the UNO shows live on the page. This is how we tested the basic motions, the line sensors and the encoders without a track. As a safety measure, the RedBot stops if no command arrives for 500 ms.

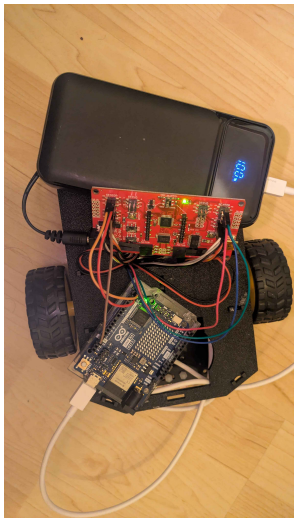


Figure 10: RedBot with the UNO R4 WiFi mounted on top, wired over serial RX/TX, and powered by a power bank.

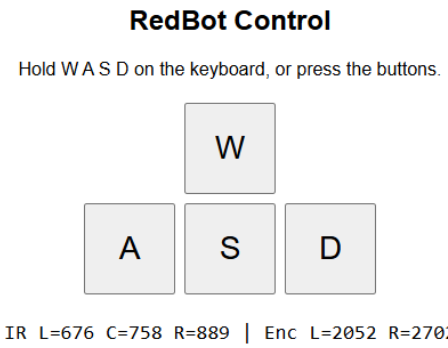


Figure 11: Web control page served by the UNO R4 WiFi, with live IR (left/centre/right) and encoder readings.

Contribution

Activity	Binay Kumar Sah	Kabir Tamari
(a) Arduino	50%	50%
(b) Prototyping	50%	50%
(c) Drone	80%	20%
(d) Ground robot	20%	80%
Report	50%	50%